aiResponseStream 178
AI workflows, memory components 185–188
API configuration 42
API keys and secrets management 285–287
API rate limiting 326
APIs, OpenAI, accessing 354
app router 45
Astra AI 20, 76–79
  backend implementation 77–78
  frontend implementation 79
  integrating streaming into 69–72
  Vercel AI SDK with 58–61
augmentation, document augmentation 222
authentication 313
  with Clerk.js 306
autoscroll 29

B

backend 37–41
background processing 326
bias, handling 17

C

caching layers 7
callback handlers 253
CCPA (California Consumer Privacy Act) 16
chain execution errors 251–254
chain-of-thought prompting 130, 151–154
  example of 151
  general methodology for creating prompts 152
ChatBubble component 100
ChatList 32–37
  component 118
ChatMessageHistory format 185
ChatOpenAI model 170
ChatPage 32
chat routes 324
ChatThread component 311
ChuckNorrisFactCard component 348
chunked transfer encoding 63
citation-based grounding 225
Clerk.js
  authentication with 306, 318
  authentication with Next.js and 277–280
  integrating authentication 359
client-side processing 63
client validation 271
code generation, defined 4
completion, defined 63
compliance and data protection 288–290
components, defined 28
composition primitives 195
compound messages 138
computational cost 143
concrete factories 75
conditional auto-scrolling 31
configuration, Next.js 45–52
console logs 237
constrained validation 144
containerization, defined 7
content enhancement, defined 4
context length limitations 192
contextual grounding 225
contextual understanding 142
control tokens 136
conversational UIs (user interfaces)
  enhancing with multimedia content 79–86
  RSCs 90–103
conversation and state management, tool and function calling 121–125
ConversationChain component 187
conversations 88
  structured data generation 113–121
CORS (cross-origin resource sharing) 39, 42, 313
cosine similarity 144, 161, 163

D

data fetching and rendering
  advanced 51
data preprocessing 13
data sensitivity 271
debug chains 253
debugging, Next.js applications 233–241
  client-server problems 237–238
  memory profiling 240
  network monitoring 240
  performance monitoring 239–241
  rendering issues 233–237
  scroll behavior problems 235
  streaming responses problems 236
  state management 239
debugging techniques 232